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STUDY PROTOCOL

An adversarial collaboration protocol for testing contrasting predictions of global neuronal workspace and integrated information theory

Lucia Melloni[1,2‡]*, Liad Mudrik[3,4‡], Michael Pitts[5‡], Katarina Bendtz[6], Oscar Ferrante[7], Urszula Gorska[8], Rony Hirschhorn[4], Aya Khalaf[9,10], Csaba Kozma[8], Alex Lepauvre[1], Ling Liu[11,12], David Mazumder[6], David Richter[13], Hao Zhou[14,15], Hal Blumenfeld[8], Melanie Boly[16], David J. Chalmers[17], Sasha Devore[2], Francis Fallon[18], Floris P. de Lange[13], Ole Jensen[7], Gabriel Kreiman[6,19], Huan Luo[11,12], Theofanis I. Panagiotaropoulos[14], Stanislas Dehaene[14,20‡], Christof Koch[21‡], Giulio Tononi[8‡]

1 Neural Circuits, Consciousness and Cognition Research Group, Max Planck Institute for Empirical Aesthetics, Frankfurt am Main, Germany, **2** Department of Neurology, New York University Grossman School of Medicine, New York, New York, United States of America, **3** School of Psychological Sciences, Tel-Aviv University, Tel Aviv, Israel, **4** Sagol School of Neuroscience, Tel Aviv University, Tel Aviv, Israel, **5** Psychology Department, Reed College, Portland, Oregon, United States of America, **6** Children's Hospital, Harvard Medical School, Boston, Massachusetts, United States of America, **7** Centre for Human Brain Health, School of Psychology, University of Birmingham, Birmingham, United Kingdom, **8** Department of Psychiatry, University of Wisconsin-Madison, Madison, Wisconsin, United States of America, **9** Department of Neurology, Yale School of Medicine, New Haven, Connecticut, United States of America, **10** Biomedical Engineering and Systems, Faculty of Engineering, Cairo University, Giza, Egypt, **11** School of Psychological and Cognitive Science, Peking University, Peking, China, **12** IDG/McGovern Institute for Brain Science at Peking University, Peking, China, **13** Donders Institute for Brain, Cognition and Behaviour, Radboud University Nijmegen, Nijmegen, The Netherlands, **14** Cognitive Neuroimaging Unit, Commissariat à l'Energie Atomique (CEA), Institut National de la Santé et de la Recherche Médicale (INSERM) U992, Gif-sur-Yvette, France, **15** State Key Laboratory of Brain and Cognitive Science, Institute of Biophysics, Chinese Academy of Sciences, Beijing, China, **16** Department of Neurology, University of Wisconsin-Madison, Madison, Wisconsin, United States of America, **17** Department of Philosophy, New York University, New York, New York, United States of America, **18** Philosophy Department, St. John's University, New York, New York, United States of America, **19** Center for Brains, Minds and Machines, Boston, Massachusetts, United States of America, **20** Collège de France, Paris, France, **21** MindScope Program, Allen Institute, Seattle, Washington, United States of America

‡ LM, LM and MP share co-first authors on this work. SD, CK and GT are co-senior authors on this work.
* lucia.melloni@ae.mpg.de

Abstract

The relationship between conscious experience and brain activity has intrigued scientists and philosophers for centuries. In the last decades, several theories have suggested different accounts for these relationships. These theories have developed in parallel, with little to no cross-talk among them. To advance research on consciousness, we established an adversarial collaboration between proponents of two of the major theories in the field, Global Neuronal Workspace and Integrated Information Theory. Together, we devised and preregistered two experiments that test contrasting predictions of these theories concerning the location and timing of correlates of visual consciousness, which have been endorsed by the theories' proponents. Predicted outcomes should either support, refute, or challenge these theories. Six theory-impartial laboratories will follow the study protocol specified here, using

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three complementary methods: Functional Magnetic Resonance Imaging (fMRI), Magneto-Electroencephalography (M-EEG), and intracranial electroencephalography (iEEG). The study protocol will include built-in replications, both between labs and within datasets. Through this ambitious undertaking, we hope to provide decisive evidence in favor or against the two theories and clarify the footprints of conscious visual perception in the human brain, while also providing an innovative model of large-scale, collaborative, and open science practice.

Introduction

Understanding how consciousness relates to neural activity in the human brain remains one of the greatest scientific challenges [1]. Phenomenologically, consciousness has been defined as *subjective experience* (or *what it is like* to perceive, feel, act or think from a first-person perspective) [2]. In human experiments, consciousness is typically inferred based on the ability to report these experiences (either to oneself or to an external observer [3]). The neural correlates of consciousness (NCCs) have been defined as the *minimal* neuronal mechanisms jointly sufficient for any specific experience [4]. Significant progress has come from focusing on the search for the NCC [5, 6]. This has led to the development of several promising scientific theories of consciousness [7].

These theories, however, have evolved independently, without cross-talk. Empirical research has focused on testing theories separately, rather than on directly contrasting them to evaluate their explanatory and predictive power [8]. Dissatisfaction with this state of affairs has inspired an alternative model of research, in which adherents of competing views work together to derive experiments that directly compare predictions [9]. Dubbed *adversarial collaboration*, the goal is to reach an agreed-upon experimental design and to document protagonists' expectations concerning the outcomes of the experiment prior to the acquisition and analysis of the data [10, 11] (for a precursor to such an approach, see [12]). This approach can be combined with pre-registration of the research protocols and methods [13].

The present study protocol describes an adversarial collaboration involving two leading theories of consciousness: *Global Neuronal Workspace* (GNW) and *Integrated Information Theory* (IIT). The status of these theories as two of the most widely-discussed and well-supported in the field has been confirmed by recent reviews, meta-analyses, and surveys [14–16]. The project will focus on conscious vision in human subjects and involve two experiments, using three complementary methods in cognitive neuroscience: functional magnetic resonance imaging (fMRI), simultaneous magnetoencephalography & electroencephalography (M-EEG) and intracranial electroencephalography (iEEG). It will include built-in replications in two forms: first, for each method, data will be acquired from two independent laboratories. Second, the data will be split in half and the second half will be used for replication of analyses. In addition, the data and analysis protocols will be openly shared for reproducibility, and large sample sizes will be attained for statistical robustness. Taken together, these open science and collaborative practices should yield reliable results that can provide substantial evidence for one or the other theory, in order to arbitrate between them. For readers who wish to review the original preregistration and its amendments, detailing the experimental protocol and analysis plans, please see the OSF preregistration (https://osf.io/mbcfy/).

Overview of the competing theories

Global neuronal workspace theory (GNW) posits that what we subjectively experience as a conscious state, at any given moment, is the global broadcasting and amplification of information [17] across an interconnected network of prefrontal-parietal areas and many distant high-level sensory cortical areas [18–20]. Unconscious processing occurs in parallel in many localized, modular circuits (e.g., in the ventral visual stream); if this processing ignites the global neuronal workspace (at about 250 ms post-stimulus presentation), information becomes conscious, being broadcasted and sustained by the workspace [18]. The workspace is constituted by a network of cortical neurons with long-range reciprocal projections to homologous neurons in other cortical areas, distributed over prefrontal (PFC), parieto-temporal and cingulate associative cortices. These neurons, mostly pyramidal cells of layers 2 and 3 (but also in layer 5), are connected through long-range excitatory axons to high-level sensory areas, allowing for flexible, domain-general amplification and distribution and exchange of information to various cognitive systems [21].

Anatomical tracer studies in non-human primates support the existence of a higher-order associative cortical network, with direct reciprocal connections between its component areas, that match the GNW core areas. Specifically, lateral prefrontal and posterior parietal cortex are connected through the superior longitudinal/arcuate fasciculus, while lateral prefrontal and superior lateral temporal cortex are connected through the extreme capsule fasciculus [22]. Parietal and temporal areas that are connected with prefrontal cortex are also heavily interconnected through the middle longitudinal fasciculus [23, 24]. Non-invasive diffusion tensor imaging studies indicate the conservation of these pathways in the human brain [22]. Importantly, these cortical areas that GNW proposes as critical for conscious information processing have been shown to comprise a highly interconnected cortical core, characterized by high information transfer efficiency [25]. This frontal-parietal-temporal network was found through graph theoretical measures to be at the top of a hierarchical cortical organization. Schematically, these areas can be viewed as being in the center of a bow-tie representation of interareal architecture and connected through feedback and feedforward loops with hierarchically lower areas in the periphery of this structure. GNW proposes that conscious access to a given information occurs when this central core network becomes invaded by neural activity coding for this information, which is achieved when this activity reaches a threshold for global ignition [26].

Integrated information theory (IIT) addresses consciousness starting from phenomenology—the existence of one's own experience, which is immediate and indubitable. As a first step, IIT identifies five properties of consciousness, which it calls "axioms", that are essential and true of every conceivable experience [27, 28]. The five axioms of IIT are intrinsic existence, composition, information, integration, and exclusion. Briefly, intrinsicality means that every experience is subjective—for the intrinsic perspective of the subject of experience, rather than for something extrinsic to it. Composition means that every experience is structured, being composed of phenomenal distinctions and relations, rather than without content. Information means that every experience is the specific way it is, rather than generic. Integration means that every experience is unitary, being irreducible to independent components, rather than multiple. Finally, exclusion means that every experience is definite—it contains what it contains, rather than having no definite border and grain.

IIT then seeks to provide an explanation for experience in physical terms, where "physical" has a precise operational definition in terms of cause-effect power—being able to take or make a difference. A physical substrate is simply a set of units that can be observed and manipulated, such as neurons that may be made to fire or not. In principle, the cause-effect